

Consensus Is Not Completion: False Convergence in Multi-Agent Discovery Workflows

Anonymous submission

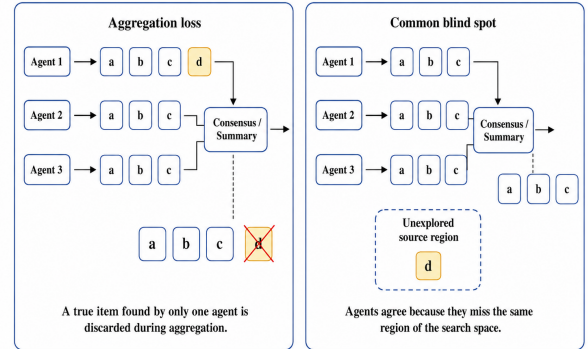
Abstract

Multi-agent language-model workflows often treat agreement, overlap, or a final summary as evidence that a task is complete. This is unsafe in closed-world discovery tasks, where the goal is to find all relevant items. We introduce *false convergence*: a workflow state in which agents report completion and exhibit high agreement, yet the final answer omits true items. We identify two mechanisms: *aggregation loss*, where minority-discovered true items are discarded during consensus or summarization, and *common blind spots*, where agents agree because they miss the same region of the search space. On a real Click repository audit, a standard LLM summarizer self-reports completion in all three seeds while retaining only 68.5–74.5% of oracle items. We argue that completion should be modeled as coverage-risk assessment under potentially correlated exploration rather than vote-counting. We implement a decoupled completion certificate with pre-audit and post-audit states: the pre-audit certificate uses only exploration logs, while holdout gain is reserved for audit and post-audit updates. Experiments on synthetic tasks and two real-repository audits show that simple stopping baselines false-certify many incomplete states. The proposed certificate avoids false certification on the current mixed-state suite, but its safe coverage is low. A first v2 offline diagnostic run shows that calibrated risk estimation can improve in-distribution risk-coverage tradeoffs, yet it does not transfer under repository/task-family holdout. A Requests online audit-controller suite over five seeds and four discovery configurations shows that targeted audit recovers missing true positives with explicit token and wall-clock accounting. A frozen staged controller evaluated on held-out Click nearly matches always-holdout recall at about one third of the audit-token cost and abstains without false certification, but safe stopping and public-benchmark validation remain open.

1 Introduction

Multi-agent language-model systems are increasingly used for tasks that require search, decomposition, and synthesis. A common design pattern is simple: multiple agents nominally inspect a task without shared intermediate answers, their outputs are aggregated by majority vote or a summarizer, and the system stops when agents agree or report high confidence. This pattern is natural. Agreement is easy to measure, and summaries provide a clean final answer.

However, agreement is not the same as completion. In closed-world discovery tasks, such as finding every affected



High agreement may reflect correlated exploration, not complete coverage.

Figure 1: Two false-convergence mechanisms. Aggregation loss drops minority true evidence during consensus or summarization; common blind spots arise when agents agree because they miss the same source region.

file location, every policy case, or every relevant document in a bounded corpus, the central question is not only whether the final answer is plausible. It is whether there remains missing mass outside the discovered set. A group of agents can converge for the wrong reason: they may share a search strategy, omit the same boundary cases, or discard minority evidence during aggregation.

This paper studies this failure mode. We call it *false convergence*: a workflow state in which the system appears complete under its own completion signals but is incomplete under a closed-world oracle. The failure is subtle because the final output can have perfect precision and high self-reported confidence. The error is not necessarily hallucination. It is often omission.

We make three contributions.

- We define false convergence in closed-world agentic discovery and distinguish two mechanisms: aggregation loss and common blind spots.
- We model completion as coverage-risk assessment rather than vote-counting over agent outputs.
- We provide a controlled empirical study showing that consensus, summarization, and raw union fail in distinct

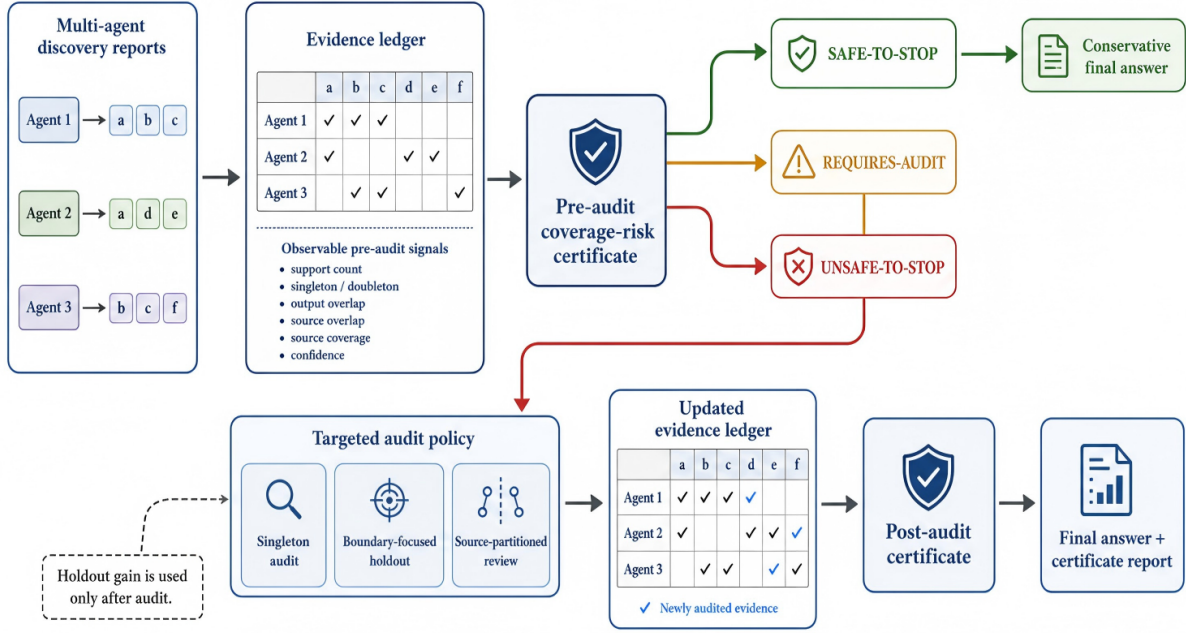


Figure 2: Decoupled completion workflow. The certificate decides whether the workflow can stop; the audit policy decides how to recover missing evidence. Holdout gain is not used as a pre-audit feature.

ways, and identify when completion-risk estimation does not transfer across repositories.

Our results do not imply that union aggregation is always better than consensus. Instead, they show that singleton evidence has two faces. Sometimes it is missing mass; sometimes it is noise. A reliable completion controller should model this distinction rather than silently deleting minority evidence or blindly accepting all unique items.

2 Related Work

2.1 Multi-Agent Consensus, Aggregation, and Calibration

Self-consistency selects a consistent answer from multiple samples (Wang et al. 2022); debate, collaboration, and role-specialized agent frameworks improve reasoning or task performance (Du et al. 2023; Chan et al. 2023; Wu et al. 2023; Chen et al. 2023; Hong et al. 2023; Li et al. 2023a; Zhang et al. 2024). Recent work also studies majority-failure auditing, selection bottlenecks, trace-level synthesis, and calibration under communication-induced correlation (Yang et al. 2026; Maryanskyy 2026; Fadnavis, Kanakaraj, and Wyss 2026; Huang et al. 2026a). Tool-using and planning agents extend LLMs through environment interaction (Yao et al. 2023b; Schick et al. 2023; Qin et al. 2023; Yao et al. 2023a; Wang et al. 2023). Benchmarks such as AgentBench, WebArena, GAIA, and SWE-bench stress interactive or software tasks (Liu et al. 2023; Zhou et al. 2023; Mialon et al. 2023; Jimenez et al. 2023). Our focus is different: aggregation is judged by closed-world recall and calibrated stopping risk, not by the quality of one selected answer.

2.2 Wide Research and Completeness Evaluation

Self-Refine, Reflexion, CRITIC, Chain-of-Verification, and related factuality checks repair or verify outputs (Madaan et al. 2023; Shinn et al. 2023; Gou et al. 2023; Dhuliawala et al. 2023; Lin, Hilton, and Evans 2021; Manakul, Liusie, and Gales 2023; Li et al. 2023b). Retrieval-augmented generation and dense retrieval reduce hallucination but still face noise, negative rejection, integration, and long-context failures (Lewis et al. 2020; Guu et al. 2020; Karpukhin et al. 2020; Nakano et al. 2021; Chen et al. 2024; Li et al. 2024; Liu et al. 2024; Asai et al. 2023; Gao et al. 2023). Closed-world discovery is also related to high-recall retrieval, fact verification, and technology-assisted review, where stopping is a coverage-estimation problem (Thorne et al. 2018; Yang et al. 2018; Grossman, Cormack, and Roegiest 2015; Cormack and Grossman 2021). Our certificate adopts this stopping perspective for correlated LLM-agent exploration. Recent broad or deep research benchmarks explicitly evaluate exhaustive information seeking and stopping uncertainty (Wong et al. 2025; Gupta et al. 2026; Rafiee et al. 2026; Huang et al. 2026b; Kim et al. 2026).

2.3 Coverage Estimation and Stopping

Our certificate uses observable missing-mass signals rather than oracle labels at decision time. Singleton counts and unseen-mass estimates are conceptually related to Good-Turing, Chao-style, capture-recapture, and ecological diversity estimators (Good 1953; Chao 1984; McAllester and Schapire 2000; Chao et al. 2014; Bron et al. 2025). Recent LLM and multi-agent uncertainty methods estimate answer risk from probabilities, samples, or semantic variation (Kuhn, Gal, and Farquhar 2023; Kadavath et al. 2022; Chen

et al. 2026; Cole et al. 2023). We study a complementary question: whether a multi-agent workflow has covered a bounded item set well enough to stop.

3 Problem Formulation

3.1 Closed-World Discovery Tasks

We consider tasks with a bounded source collection D and a finite oracle set Y^* of target items. A run returns a set $\hat{Y}$. We evaluate:

$$\text{Recall}(\hat{Y}) = \frac{|\hat{Y} \cap Y^*|}{|Y^*|}, \quad (1)$$

$$\text{Precision}(\hat{Y}) = \frac{|\hat{Y} \cap Y^*|}{|\hat{Y}|}. \quad (2)$$

Examples include code dependency audits, policy-case discovery, high-recall document review, and bounded repository analysis. Unlike open-ended question answering, a plausible final answer is insufficient: the system must decide whether it has found the complete set.

3.2 Multi-Agent Completion Signals

A typical workflow runs k agents $A_1, \dots, A_k$, each producing an item set Y_i , a completion flag c_i , and a confidence score q_i . Aggregation then produces a final set Y_{agg} . Common aggregation rules include:

- **Majority consensus:** keep items reported by at least two agents.
- **Standard summarization:** ask a summarizer to produce a clean final answer from the agent reports.
- **Raw union:** keep every unique item reported by any agent.
- **Union-preserving summarization:** instruct a summarizer to preserve all unique reported items unless malformed or contradicted.

Systems often treat high pairwise overlap, high mean confidence, and affirmative self-reported completion as stopping evidence. We argue that these are not completion certificates.

3.3 False Convergence

Let S be a multi-agent workflow state with k agents, an aggregation rule g , and a target recall threshold θ . We say S exhibits false convergence under g when:

1. agents self-report completion with high confidence;
2. agent outputs exhibit high overlap;
3. the aggregated answer $g(Y_1, \dots, Y_k)$ has recall below θ .

Our fixed evaluation protocol uses $\theta = 0.95$, confidence threshold 0.80, and overlap threshold $\tau = 0.70$. Holdout recovery is a diagnostic and audit mechanism, not part of the definition. We also report near-positive and mechanism-positive cases when the missing items are few but diagnostically important.

3.4 Aggregation Loss

Aggregation loss occurs when a true item appears in at least one agent report but is removed by consensus-style aggregation or standard summarization. This is not a search failure: the evidence exists in the reports. The failure happens at the final synthesis step.

Formally, an oracle item $y \in Y^*$ is an aggregation-loss item if:

$$y \in \bigcup_i Y_i \quad \text{and} \quad y \notin Y_{\text{agg}}. \quad (3)$$

When y appears in only one agent report, it is a singleton true positive.

3.5 Common Blind Spots

Common blind spots occur when multiple agents agree because they share a search failure. In this case, the missing item is absent even from the raw union:

$$y \in Y^* \quad \text{and} \quad y \notin \bigcup_i Y_i. \quad (4)$$

Raw union and union-preserving summarization cannot recover such items because there is no reported evidence to preserve. The mitigation must change the search or auditing process, for example through boundary-focused holdout review.

3.6 Singletons as Missing Mass or Noise

Singletons are ambiguous. A singleton may be a hard-to-find true positive, or it may be a false positive. Consensus implicitly treats singletons as noise. Raw union implicitly treats them as valid evidence. Both assumptions are unsafe. The right action is often to preserve singleton evidence for audit.

4 Evidence-Preserving Staged Audit Controller

We propose a staged audit controller for deciding whether a multi-agent discovery workflow can safely stop or should preserve evidence for further review. A lightweight completion certificate is the controller’s risk-triage module. It is not an oracle: it does not use ground truth and does not decide final item labels. Instead, it maps observable workflow signals to a stopping or escalation judgment.

4.1 Observable Signals

The pre-audit certificate takes as input only exploration-stage reports $Y_1, \dots, Y_k$, confidence values, and source metadata. It does not use oracle labels or holdout gain. We compute item counts $n(y)$ and frequency counts $f_r = |\{y : n(y) = r\}|$. The Chao-style unseen estimate is

$$\hat{U}_{\text{Chao}} = \begin{cases} f_1^2 / (2f_2), & f_2 > 0, \\ f_1(f_1 - 1) / 2, & f_2 = 0, \end{cases} \quad (5)$$

with missing ratio $m_{\text{Chao}} = \hat{U}_{\text{Chao}} / (|\bigcup_i Y_i| + \hat{U}_{\text{Chao}})$. We also compute Good–Turing missing mass $m_{\text{GT}} = f_1 / \sum_y n(y)$, mean output Jaccard, mean source overlap, source coverage, and an effective exploration size $k_{\text{eff}} = k / (1 + (k - 1)\rho)$, where ρ is mean output Jaccard.

4.2 Certificate Rule

The implemented rule computes a continuous risk score

$$R = r_{\text{conf}} + .25 \min(1, s/0.20) + .22 \min(1, m_{\text{Chao}}/0.10) + .20 \min(1, m_{\text{corr}}/0.15) + r_{\text{src}} + r_{\text{boundary}} + r_{\text{audit}}, \quad (6)$$

clipped to $[0, 1]$, where $s = f_1/|\cup_i Y_i|$ and $m_{\text{corr}} = \min(1, m_{\text{Chao}} \cdot k/k_{\text{eff}})$. The confidence penalty r_{conf} is 0.12 when mean confidence is below 0.80. The source penalty is proportional to missing source coverage below 0.30. Boundary risk adds 0.12 when mean overlap is at least 0.95 on a boundary-sensitive task. Post-audit unresolved singleton evidence adds 0.12.

The label is SAFE-TO-STOP only when confidence is high, $m_{\text{corr}} \leq 0.05$, singleton evidence is absent or resolved by audit, and $R < 0.25$. It is UNSAFE-TO-STOP when $R \geq 0.60$ or $m_{\text{corr}} \geq 0.20$; otherwise it returns REQUIRES-AUDIT. These thresholds are fixed in code and written to the result log.

4.3 Certificate Outputs

The certificate returns one of three labels:

- **SAFE-TO-STOP:** observable risk is low enough that the workflow may publish the current conservative answer.
- **UNSAFE-TO-STOP:** observable risk suggests likely missing mass; the workflow should continue searching or run a stronger audit.
- **REQUIRES-AUDIT:** the current answer may be complete, but singleton evidence, noisy union items, or weak independence prevents clean certification.

In the current experiments, the certificate is deliberately conservative. It is designed to catch high-risk states rather than to maximize automatic stopping. We therefore report false-certification rate and safe coverage separately.

4.4 Evidence-Preserving Audit Policy

When the certificate returns UNSAFE-TO-STOP or REQUIRES-AUDIT, the workflow needs an audit policy. We use an evidence-preserving policy that separates finalization from evidence retention: consensus items form a conservative answer, singleton evidence is retained for review, and high-agreement boundary-risk states trigger holdout or source-focused audit. This policy is the audit response to a risky certificate state, not the certificate itself.

The audit policy handles the two failure mechanisms differently. For aggregation loss, it preserves and audits singleton or minority evidence instead of deleting it during summarization. For common blind spots, where no current agent reported the missing item, it triggers boundary-focused holdout or source-focused review. In our experiments, singletons are not automatically accepted; they are retained and audited. High agreement is not automatically trusted; it can trigger a common-blind-spot check. The full procedural listing is in the supplementary appendix.

The frozen staged controller used after the Requests development experiment therefore applies a simple order: conservative consensus, singleton verifier, source-partitioned

review when singleton mass, union gap, or source-family gaps remain, and abstention if escalation was needed. This makes recovery and certification separate: the controller may recover true positives while still refusing to certify completion.

5 Experimental Setup

5.1 Research Questions

We ask:

- **RQ1:** Can consensus-style aggregation produce false completion signals in closed-world discovery tasks?
- **RQ2:** Are failures caused by aggregation loss, common blind spots, or both?
- **RQ3:** Does raw union solve the problem, or does it introduce a precision cost?
- **RQ4:** Can a coverage-risk certificate identify unsafe stopping states, and when can an audit policy recover missing items?

5.2 Tasks

We use four bounded discovery task families: two synthetic families designed to isolate mechanisms, and two real-repository audits used to test whether the failure appears beyond constructed examples.

T1: Code dependency audit. Agents inspect a synthetic AcmePay migration repository and must find every code or configuration location that resolves to AcmePay v1 through direct calls, routing configuration, registry indirection, fallback queues, or replay scripts. The hard oracle has 35 items.

T2: Policy document search. Agents inspect naturalistic incident notes, changelogs, appendices, and policy memos. They must find all cases whose state, flow, and service-resolution chain imply AcmePay v1 handling. The oracle has 30 items. Boundary cases require following service catalog and adapter registry resolution rather than relying on surface service names.

T3: Real-repository Click deprecation audit. Agents inspect a fixed snapshot of the public Click repository and must find line-level locations that belong to a deprecated API surface audit. The oracle has 149 items spanning implementation, warning emission, help-message formatting, parameter behavior, tests, documentation, and bounded changelog entries. Unlike T1/T2, T3 uses a real repository and is used as a stronger validation of aggregation behavior rather than as a strict false-convergence construction.

T4: Real-repository Requests TLS audit. We additionally instantiate a second real-repository task family from a fixed snapshot of the public Requests repository. The task asks agents to find line-level items related to TLS certificate verification, CA bundle resolution, client certificates, SSL error handling, tests, test infrastructure, and user documentation. The oracle has 304 items. T4 is used to test whether the real-repository result depends on Click or deprecation-surface search.

5.3 Runs and Aggregation Policies

Each seed contains three G3 agents. Where available, we also run an independent G6 holdout scout. To test whether the certificate simply always refuses to stop, we construct mixed evaluation states from saved logs: each single agent, each two-agent union, the three-agent pre-audit union, and the post-holdout state when available. This yields complete, near-complete, and incomplete states without generating new oracle-dependent agent outputs. We compare:

- majority consensus;
- blind standard summarization;
- raw union;
- blind union-preserving summarization;
- holdout scout;
- self-reported completion, confidence-only, overlap-only, no-new-item, Chao-only, always-unsafe, always-holdout, and the proposed certificate.

The standard and union-preserving summarizers are blind to oracle and scoring files. The pre-audit certificate uses only exploration outputs and metadata; holdout gain is used only after audit. Oracle is used only for offline metrics.

5.4 Closed-Loop Evaluation and Data Volume

The experiment is designed as a closed loop rather than a collection of isolated case studies. Each task has a bounded source collection and oracle. Each run produces agent item sets and metadata. Aggregation policies produce candidate answers. The pre-audit certificate decides whether the state is safe, unsafe, or requires audit without access to oracle labels or holdout gain. Audit policies then update the evidence ledger, and final recall, precision, false-certification rate, audit recovery, and cost metrics are computed offline. This loop lets us evaluate both failure diagnosis and the downstream control decision.

Table 1 summarizes the scale used in the current paper. The online blind LLM evidence covers three-agent seeds on two synthetic task families and two real-repository audits. The v1 certificate evaluation uses 96 saved mixed states from these logs. The v2 diagnostic expands the logging pipeline to 2,970 deterministic offline runs and 2,340 pre-audit states; it is used for calibration stress testing, not as a substitute for online blind agent validation. The Requests online audit-controller suite adds five seeds, four discovery configurations per seed, 60 blind discovery calls, and 45 online audit calls with token and wall-clock accounting.

6 Results

6.1 Mechanism Evidence

The six mechanism slices show the two failure modes. T1 seed01 and T2 seed01 show aggregation loss: true items were present in minority reports but dropped by consensus and standard summaries. T2 seed02 and seed03 show common blind spots: all three G3 agents agreed on the same incomplete 28-item set, and raw union could not recover the missing cases. T1 seed02 and seed03 are precision-cost controls: consensus is complete, while raw union preserves false positive singletons.

6.2 Aggregation Policy Comparison

Table 2 shows the method-level comparison. Consensus and standard summarization can produce high-precision false stops. Raw union and union-preserving summarization recover aggregation-loss cases, but incur precision cost in T1 seed02/03 and cannot recover T2 common blind spots. The evidence-preserving audit policy recovers missing items when holdout evidence is available and marks singleton-noise controls as requiring audit rather than silently accepting false positives.

6.3 Real-Repository Summarizer False Stops

T3 evaluates whether the aggregation-stage failure appears in a real repository task rather than only in constructed mechanism slices. We give the same G3 aggregation packet to two blind LLM summarizers. The *standard* summarizer is instructed to produce a concise, high-precision final answer from the three agent reports. The *union-preserving* summarizer is instructed to preserve all unique reported items unless malformed or contradicted.

Table 3 shows a stable pattern across three blind seeds. The standard summarizer reports completion with confidence between 0.93 and 0.96, but its recall is only 0.685–0.745. Its precision is nearly perfect, so the failure is not hallucination; it is omission. The union-preserving summarizer removes the false stop in all three seeds, reaching 0.993–1.000 recall, but precision falls to 0.639–0.744. This supports two claims. First, much of the missing evidence was already present in the agent reports and was discarded during aggregation. Second, naive union preservation is diagnostic but not a complete solution because it preserves noise along with missing true items.

6.4 Second Real-Repository Task Family

T4 evaluates a different real repository and audit surface: TLS and certificate handling in Requests. Unlike T3, this task is larger and noisier. Table 4 shows that the failure shifts from mostly aggregation loss to mixed search-coverage and aggregation risk. Majority consensus is incomplete in all three seeds, with recall between 0.678 and 0.701. Raw union improves recall to 0.829–0.859, showing that some omitted evidence is present in minority reports, but raw union remains below the 0.95 completion threshold in every seed. Thus T4 contains candidate-pool missing mass that union aggregation cannot recover.

The standard summarizer also false-stops in all three seeds, with recall 0.368–0.648 despite self-reported completion. The completion certificate marks all three seeds UNSAFE-TO-STOP. This supports the broader claim that a completion controller must estimate coverage risk, not merely summarize or vote over the current reports. T4 should not be read as a direct copy of the T3 mechanism; instead, it is a harder boundary case showing that real-repository discovery can combine aggregation loss, noisy singleton evidence, and insufficient search coverage.

6.5 Decoupled Certificate and Stopping Baselines

We then evaluate the decoupled v1 certificate on 96 saved states constructed from existing runs: 86 pre-audit states

Evidence source	Task families	Oracles	Seeds	Runs/states	Log rows	Role
Mechanism slices	T1, T2	35/30	3 each	6 G3 packets	–	mechanisms
Real-repo blind runs	T3, T4	149/304	3 each	6 G3 packets	–	validation
Requests online audit	T4	304	5	60 discovery / 45 audit calls	3.90M tokens	policy eval
v1 certificate suite	T1–T4	mixed	mixed	96 states	–	stopping baselines
v2 offline diagnostic	T2, T3, T4	mixed	5 each	2,970 runs / 2,340 states	1,572,201	calibration stress test

Table 1: Current experimental scope. The v2 row is a deterministic offline diagnostic with complete candidate/action logging; the Requests online row is a blind online audit-controller evaluation on one real repository task family.

Case	Method	Found	TP	FP	Recall	Status
T1 seed01	consensus / standard	22	22	0	0.629	false stop
T1 seed01	raw union / union-preserving	35	35	0	1.000	complete
T1 seed01	audit policy	35	35	0	1.000	verified
T1 seed02	consensus / standard	35	35	0	1.000	complete
T1 seed02	raw union / union-preserving	39	35	4	1.000	precision cost
T1 seed02	audit policy	35	35	0	1.000	requires audit
T1 seed03	consensus / standard	35	35	0	1.000	complete
T1 seed03	raw union / union-preserving	40	35	5	1.000	precision cost
T1 seed03	audit policy	35	35	0	1.000	requires audit
T2 seed01	consensus / standard	28	28	0	0.933	false stop
T2 seed01	raw union / union-preserving	30	30	0	1.000	complete
T2 seed01	audit policy	30	30	0	1.000	verified
T2 seed02	consensus / raw union	28	28	0	0.933	common blind spot
T2 seed02	holdout / audit policy	30	30	0	1.000	verified
T2 seed03	consensus / raw union	28	28	0	0.933	common blind spot
T2 seed03	holdout / audit policy	30	30	0	1.000	verified

Table 2: Method comparison on current case slices. Grouped rows combine methods with identical scores in the current experiments.

and 10 post-audit states. Oracle labels are used only for evaluation; a state is safe when recall is at least 0.95. The suite contains 44 safe and 52 unsafe states. Table 5 shows that simple stopping signals false-certify many incomplete states: self-reported completion and confidence-only stopping each false-certify 35 states, Chao-only false-certifies 13, and overlap-only false-certifies 10. The proposed certificate certifies one post-audit state and makes no false certifications, but its safe coverage is only 0.023. Thus the current rule is a conservative risk detector, not a solved stopping policy. Its continuous risk score has AUROC 0.515 and AUPRC 0.589 for unsafe-state detection, indicating that stronger correlation and coverage features are still needed.

Rule	Stop	False	FCR	Safe Cov.
Self-report	70	35	.500	.795
Confidence	70	35	.500	.795
Overlap	15	10	.667	.114
No-new-item	22	10	.455	.273
Chao-only	26	13	.500	.295
Raw union	86	49	.570	.841
Always unsafe	0	0	.000	.000
Always holdout	10	3	.300	.159
Ours v1	1	0	.000	.023

Table 5: Stopping evaluation on 96 mixed saved states. FCR is false certification rate among stopped/certified states. The proposed certificate is safe but too conservative in this version.

6.6 Transferability Audit of v2

We freeze v1 and the v2 offline diagnostic before further inspection. A leakage audit finds no direct use of oracle labels, holdout gain, task id, repository id, or post-audit cost columns in the declared v2 feature set. However, a metadata-only shortcut probe has high in-distribution AUROC, and leave-one-repository-out validation fails to certify any held-out repository under the same calibration rule: AUROC is

Seed	Policy	Conf.	Found	TP	FP	Recall	Precision	False Stop
01	standard summarizer	0.930	105	104	1	0.698	0.990	yes
01	union-preserving summarizer	0.910	199	148	51	0.993	0.744	no
02	standard summarizer	0.960	111	111	0	0.745	1.000	yes
02	union-preserving summarizer	0.950	233	149	84	1.000	0.639	no
03	standard summarizer	0.930	102	102	0	0.685	1.000	yes
03	union-preserving summarizer	0.840	232	149	83	1.000	0.642	no
Mean	standard summarizer	—	—	—	—	0.709	0.997	3/3
Mean	union-preserving summarizer	—	—	—	—	0.998	0.675	0/3

Table 3: T3 real-repository summarizer comparison. The oracle contains 149 line-level Click deprecation-audit items. Standard summarization is high-precision but repeatedly false-stops; union-preserving summarization recovers recall but incurs a substantial precision cost.

Seed	Cons. Rec.	Std. Rec.	Union Rec.	Union Prec.	Holdout Rec.	Cert.
01	0.701	0.368	0.859	0.661	0.750	unsafe
02	0.678	0.622	0.845	0.668	0.688	unsafe
03	0.678	0.648	0.829	0.676	0.763	unsafe
Mean	0.685	0.546	0.844	0.668	0.734	3/3 unsafe

Table 4: T4 Requests TLS/certificate audit on a second real repository. Raw union improves over consensus but remains incomplete, indicating missing mass outside the G3 candidate pool. Standard summarization false-stops in all three seeds.

.591 for Click, .546 for policy docs, and .695 for Requests, with zero certified held-out states. Thus v2 remains an offline diagnostic rather than the main method; the full leakage and generalization table is in the supplementary appendix.

6.7 Online Audit-Controller Evaluation

We run the Requests TLS online audit-controller suite with a fixed model, repository commit, prompt family, budget, and seeds 04–08. Each seed has four blind discovery configurations: homogeneous, prompt-diverse, source-partitioned, and independent-context. We then evaluate seven audit policies using only pre-audit discovery logs to build audit queues. Online singleton and random verifiers review candidate line snippets; boundary-focused holdout agents run a separate blind search. Oracle labels are used only after audit evidence is written.

Policy	Pre R	Post R	Prec.	Rec. TP	Intro. FP	Tok.
No audit	.491	.491	.829	0.0	0.0	0k
Random holdout	.491	.554	.755	18.9	11.9	17k
Singleton audit	.491	.719	.725	69.1	38.7	12k
Boundary holdout	.491	.694	.755	61.7	24.6	71k
Source-partitioned review	.491	.733	.730	73.3	38.5	55k
Risk-triggered audit	.491	.733	.713	73.3	45.5	83k
Always holdout	.491	.753	.706	79.5	51.1	163k

Table 6: Online Requests TLS audit-policy results averaged over 20 seed/configuration states. Pre R is the conservative consensus answer; union items are retained as an evidence queue and enter the final answer only after audit. Tokens are additional audit tokens, not discovery tokens.

Table 6 and Figure 3 show that auditing materially improves recall, but does not solve completion. Singleton

audit is the cheapest strong recovery mechanism because many missed true items are already present as minority evidence. Source-partitioned review and risk-triggered audit reach similar mean recall, while always-holdout buys the highest recall at much higher cost. All policies remain below the 0.95 completion threshold, so the controller should continue to report UNSAFE-TO-STOP rather than certify completion.

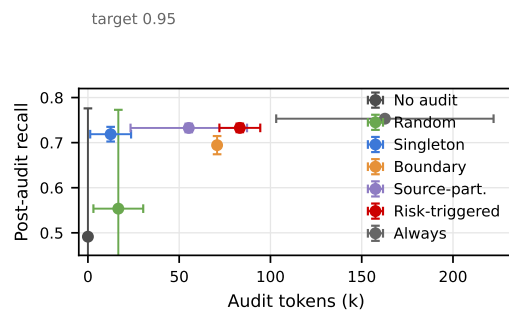


Figure 3: Online audit-policy recall/cost tradeoff on Requests TLS. Points and error bars show means and standard deviations over 20 seed/configuration states.

We then freeze the staged controller and evaluate it on held-out Click online runs with three new seeds and three discovery configurations. The staged controller reaches mean recall .761 versus .768 for always-holdout, while using 56k rather than 177k additional audit tokens. However, it abstains on every held-out state, yielding zero false certifications but also zero safe coverage. Thus the held-out

result supports recall–cost recovery and reliable abstention, not automatic safe stopping.

7 Discussion and Limitations

Consensus can fail for two opposite reasons. It can be too conservative when a minority report contains real evidence. It can also be overconfident when agents appear to share exploration patterns associated with the same blind spot. Thus agreement is evidence of shared sampling behavior, not necessarily evidence of low missing mass. Raw union diagnoses aggregation loss, but can preserve singleton false positives and cannot recover unreported items. Completion should therefore be treated as coverage-risk assessment using singleton status, overlap, source coverage, and boundary-risk signals.

The evidence supports mechanism-level claims, real-repository false stops, and a proof-of-concept coverage-risk certificate, not a general proof of safe completion. v2 passes direct feature-column leakage auditing but fails leave-one-repository/task-family validation, so it is not promoted to the main method. The online audit-controller evidence now includes Requests as a development task and Click as a held-out repository test, but it is still not a benchmark sweep. Broader claims require more repositories, independent oracle review, second-model validation, and public benchmark validation; SeekerGym is scaffolded but not run.

8 Conclusion

Consensus is not completion. We identified aggregation loss and common blind spots as distinct reasons why high agreement can still omit true items. The current online audit-controller results show that targeted audit can recover missing true positives with measured cost, but they also show that recovery is not yet completion. Broader external benchmarks are required before claiming a general cost-effective solution.

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